

Chris Melski

SENIOR TEST MANAGER | SDET

 c_melski@yahoo.com |  <https://cmelski.github.io/online-portfolio/> |  <https://www.linkedin.com/in/chrismelski/>
|  <https://github.com/cmelski>

Professional Summary

Results-driven Test Manager and SDET with 20+ years of global experience across QA strategy, test automation, and requirements management within FX Trading, Telecom, and eCommerce domains. Skilled in developing end-to-end test automation frameworks and leading large, distributed QA teams to deliver high-quality, compliant systems. Combines a strong hands-on technical foundation with strategic test leadership, ensuring quality governance, automation efficiency, and business alignment from requirements through production release.

Career Highlights

Test Management & QA Governance

- Directed end-to-end QA delivery for BARX FX e-Trading, covering pricing, order execution, risk checks, booking, and regulatory controls.
- Led global QA teams across Singapore, Pune, Prague, and London; standardized FX test coverage, automation, and reporting practices.
- Defined and enforced QA strategies including test planning, defect triage, and risk-based coverage.
- Leveraged JIRA/Xray to define and track KPIs around regression coverage, execution time, and defect leakage. I used this data to identify bottlenecks and then optimized the regression suite by prioritizing high-risk scenarios and removing low value tests, reducing regression execution time by ~80%.
- Managed SIT/UAT execution and sign-off with trading, sales, compliance, and technology stakeholders.
- Championed Agile QA transformation, embedding automation into sprint delivery for FX platforms and improving release predictability.

SDET & Test Automation Engineering

- Designed and implemented automation frameworks using Python (pytest, Playwright) for FX trading applications.
- Automated API, database, and UI testing across pricing engines, execution services, and downstream trade processing, integrated into CI/CD pipelines.
- Implemented BDD frameworks (Cucumber) aligned to FX user stories for clearer business and technology alignment.
- Conducted automated trade and data validation across FX trading systems, APIs, and databases.
- Built reusable test utilities for market data setup, trade simulation, log analysis, and results validation.
- Built numerous AI-Driven test automation frameworks covering test case generation, coverage reporting, and self-healing selector logic.

Business Analysis & Requirements Gathering

- Partnered with FX traders, sales, risk, compliance, and developers to define and test requirements for trading and regulatory changes.
- Produced functional specifications, user stories, and process flows covering pricing, execution, and post-trade workflows.
- Drove traceability from FX business requirements to automated test coverage, improving audit readiness and reducing production risk.

Key Skills

- **Test Management:** Test Strategy | QA Governance | SIT/UAT | Defect Management | Risk Assessment | Reporting | Team Leadership | Stakeholder Engagement/Management
- **SDET & Automation Engineering:** Python (pytest, unittest) | Playwright | Cypress | Selenium | BDD (Cucumber) | REST API | SQL | CI/CD | GitHub | GitHub Actions | Jenkins | Azure Devops | FIX Protocol | JavaScript | Java | VBA | Linux
- **Business Analysis:** Requirements Analysis | User Story Mapping | Process Modeling | Test Traceability | UAT Coordination | Root Cause Analysis
- **Tools & Tech:** JIRA | Confluence | Postman | PyCharm | VS Code | MySQL | PostgreSQL | Oracle | Pandas | NumPy | Matplotlib | Excel Macros | Github Copilot
- **Domains:** FX Trading | Algorithmic Trading | eCommerce | Billing | POS | Inventory Systems
- **Soft Skills:** Communication | Problem-Solving | Negotiation | Adaptiveness

Professional Experience

Senior Test Manager | SDET Lead - Barclays, Singapore (2007 – 2024)

- Led test automation initiatives for BARX FX e-Trading platforms, covering spot, forwards, swaps, and NDF workflows, integrating Python Playwright regression suites into CI/CD pipelines.
- Drove test planning and execution across FX cash and derivatives, ensuring global test alignment across pricing, execution, STP, and post-trade flows.
- Managed and mentored a team of QA engineers and SDETs, establishing automation-first testing strategies for low-latency trading systems.
- Oversaw requirements validation and regulatory change testing (e.g., MiFID II, Dodd Frank, trade reporting), ensuring compliance and audit transparency.
- Conducted root cause analysis for production trading incidents (pricing discrepancies, execution failures, booking breaks), implementing preventive test coverage improvements.
- Acted as the bridge between QA, Development, and FX Trading, Sales, and Risk stakeholders, ensuring end-to-end delivery quality across the trade lifecycle.

QA Consultant (Independent) – Test Automation and Framework Development (2024 – Present)

- **AI-Driven Playwright Test Framework** - Built a Python/Playwright framework combining YAML-driven test specs, strict POM design, and AI-assisted test scenario generation.
- **Automated Jira Defect Integration** - Integrated Jira REST APIs into a Playwright/pytest framework via custom hooks; auto-creates or links defects on failure and attaches screenshots and Allure logs.
- **AI-Assisted Log Analyzer** - Developed a Python tool to classify pytest log failures using AI, outputting categorised error types, confidence levels, and stack traces to eliminate manual log review.
- **AI Self-Healing Selector Framework** - Implemented a GPT-driven fallback locator strategy (CSS, XPath, data-testid) within Playwright to auto-recover from UI changes, reducing test flakiness and maintenance.
- **Estate Management Web Application** - Built a full-stack Flask/PostgreSQL application with a vanilla JS frontend; delivered data-driven end-to-end test coverage using pytest (API) and Playwright (UI).
- **Config-Driven API Health Checker** - Developed a Python tool with strict config validation, structured logging, and clean error handling to monitor API endpoint health across environments.
- **Server Log Analysis Tool** - Built a Python log parser to compute new-order request rates and failed-order classifications per client, with CLI argument parsing, input validation, and CSV reporting.
- Project examples on **GitHub** - <https://github.com/cmelski>

Test Lead - Gen-i, Wellington, NZ (2006 – 2007)

- Defined and managed test strategy and requirements validation for helpdesk software implementation.
- Facilitated UAT sessions, ensuring business sign-off aligned with user expectations.

Test Engineer - Bell, Toronto, Canada (2004 – 2006)

- Supported billing system migration, combining test execution with requirements traceability.
- Developed reusable test assets and early automation scripts for regression cycles.

Business Analyst - ReserveAmerica, Mississauga, Canada (Jan 2003 – Dec 2003)

- Gathered requirements, created functional specifications, and supported testing for e-commerce POS and Ticketing systems.

Java Developer | QA Analyst - SIVCO Inc, Calgary, Canada (2002 – 2003)

- Built Java solutions for an Inventory Tracking System and introduced formal QA processes into SDLC.

Education

- Post Graduate Programming Diploma, ITI, Toronto, Canada (Hon) – 2001
- Bachelor of Business Admin (Marketing), Cape Breton University, Sydney, Canada (Hon) - 1998

Professional Development

- Azure Devops Crash Course – Build CI/CD Release Pipelines (Feb 2026), Cypress – Modern Automation Testing + Frameworks (Sep 2025), Playwright Python Automation Testing (Jun 2025), API Testing with Python and PyTest (Jan 2025), [PCEP-30-02] PCEP™ - Certified Python Programmer (Oct 2024), 100 Days of Code: Python Pro Bootcamp (Jun 2024), Learn Cucumber BDD with Java – Masterclass Selenium Framework (Apr 2024)